

Statistical modeling

Module 11

Marc-Olivier Boldi

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Generalized Linear Models

Regression and logistic regression belongs to a larger family of models called Generalized Linear Models (GLM). They can be used the same way and the choice mainly depends on the type of the response y .

- ▶ Regression: y is numerical, normally distributed (symmetric and continuous or when discontinuity is not important).
- ▶ Logistic: y is binary (0/1).
- ▶ Poisson: y is a count (0, 1, 2, etc.).
- ▶ Gamma: y is positive (asymmetric distribution with on $[0, \infty[$).

Other models have been developed depending on the specific application.

GLM: count data

Example: Medical care demand study: y is the number of visits to a physician office ofp as a function of

- ▶ Number of hospital stays: `hosp`, x_1 , num.
- ▶ Health: `health`, x_2 , (average/excellent/poor).
- ▶ Number of chronic conditions: `numchron`, x_3 , num.
- ▶ gender: x_4 (female/male)
- ▶ Number of years of education: `school`, x_5 , num.
- ▶ private insurance: `privins`, x_6 (no/yes)

Source: <https://cran.r-project.org/web/packages/pscl/vignettes/countreg.pdf>

GLM: count data

For a set of variables x , the linear predictor is (like for the logistic regression),

$$\eta(x) = b_0 + b_1x_1 + \cdots + b_px_p$$

Given x , the response $y(x)$ is assumed to be distributed according to a Poisson distribution with mean $\lambda(x)$,

$$y(x) \sim \text{Pois}(\lambda(x)).$$

In details,

$$P(y(x) = k) = \frac{\lambda(x)^k}{k!} e^{-\lambda(x)}, \quad k \geq 0.$$

The mean $\lambda(x)$ and the linear predictor $\eta(x)$ are linked by the log-link function

$$\eta(x) = \log \lambda(x)$$

GLM: count data

With R, the model is fitted as

```
> mod <- glm(ofp ~ hosp + health + numchron + gender + school + privins, family=poisson, data=dt)
> summary(mod)
[...]
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.028874	0.023785	43.258	<2e-16 ***
hosp	0.164797	0.005997	27.478	<2e-16 ***
healthexcellent	-0.361993	0.030304	-11.945	<2e-16 ***
healthpoor	0.248307	0.017845	13.915	<2e-16 ***
numchron	0.146639	0.004580	32.020	<2e-16 ***
gendermale	-0.112320	0.012945	-8.677	<2e-16 ***
school	0.026143	0.001843	14.182	<2e-16 ***
privinsyes	0.201687	0.016860	11.963	<2e-16 ***

```
---
[...]
```

GLM: count data

The estimated linear predictor is

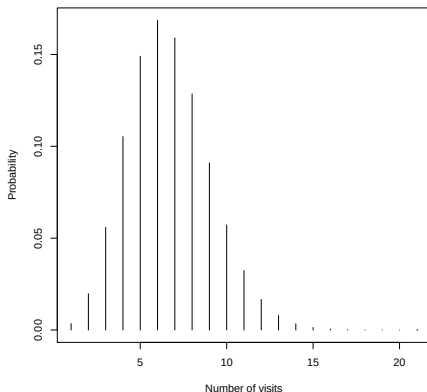
$$\begin{aligned}\hat{\eta}(x) = & 1.03 + 0.16x_1 - 0.36\delta_{\text{excl}}(x_3) + 0.25\delta_{\text{poor}}(x_3) \\ & + 0.15x_4 - 0.11\delta_{\text{male}}(x_5) + 0.03x_6 + 0.20\delta_{\text{yes}}(x_7)\end{aligned}$$

```
> dt[1,c(6,7,8,15,17,20)] ## x
  hosp health numchron gender school privins
1    1 average         2  male      6      yes
> predict(mod, type="link")[1] ## eta(x)
1
1.733175
> predict(mod, type="response")[1] ## y(x)
1
5.658592
```

The prediction can be rounded to 6 in practice.

GLM: count data

In more details, the model tells us that the number of visits of p for the population of subjects with characteristics x (previous slide) is distributed as a Poisson distribution with mean 5.66, that is



GLM: gamma distribution

Example: Insurance study (Sweden 1977). The response y is the total value payment `Payment`. It should be explained as function of

- ▶ The car model `Make`: x_1 , categorical with 9 levels,
- ▶ The number of insured in policy-years `Insured` (in log-scale): x_2
- ▶ The bonus claim `Bonus`: x_3 , (Y/N).

Source: <http://people.stat.sc.edu/Hitchcock/stat599chap7Rcode.txt>

GLM: gamma distribution

For a set of variables x , the linear predictor is (like always)

$$\eta(x) = b_0 + b_1x_1 + \cdots + b_px_p$$

Then, given x , the response $y(x)$ is assumed to be distributed as a Gamma distribution with scale $\theta(x)$ and shape r

$$y(x) \sim Ga(\theta(x), r).$$

In details, the density function of y given x is

$$f(y; \theta(x), r) = \frac{1}{\Gamma(r)\theta(x)^r} y^{r-1} e^{-y/\theta(x)}.$$

GLM: gamma distribution

The mean and variance of y given x are

$$E(y) = r\theta(x), \quad \text{Var}(y) = r\theta(x)^2.$$

The link between $\eta(x)$ and $\theta(x)$ is the inverse:

$$\eta(x) = 1/\theta(x).$$

The shape parameter r is estimated as a constant (independent of x).

GLM: gamma distribution

The fitted model is

```
> mod <- glm(Payment ~ log(Insured) + Make + Bonus, family = Gamma, motor)
> summary(mod)
```

[...]

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	6.191e-06	8.302e-07	7.456	1.08e-12	***
log(Insured)	-5.532e-07	1.117e-07	-4.951	1.27e-06	***
Make2	7.127e-06	2.114e-06	3.371	0.000852	***
Make3	9.335e-06	2.534e-06	3.684	0.000275	***
Make4	8.831e-06	2.691e-06	3.282	0.001159	**
Make5	1.040e-05	2.775e-06	3.747	0.000217	***
Make6	7.292e-06	2.102e-06	3.470	0.000602	***
Make7	1.836e-05	4.317e-06	4.253	2.87e-05	***
Make8	2.583e-05	6.086e-06	4.244	2.97e-05	***
Make9	-1.148e-06	7.221e-07	-1.590	0.112887	
Bonus	1.378e-07	3.531e-08	3.904	0.000118	***

(Dispersion parameter for Gamma family taken to be 1.126119)

GLM: gamma distribution

The prediction is the mean, that is

$$\hat{y}(x) = \hat{\mu} / \hat{\eta}(x).$$

In the example, the prediction is

```
> motori[1,c(3:5,7)] ## x
  Bonus Make Insured Payment
1      1      1 455.13 392491
> 1.126/predict(mod, type="link")[1] ## y.hat
      1
382634.1
```

because of the estimate of the shape parameter $\hat{\mu} = 1.126$.

Note: in practice, transforming the response y to its log and using a regression is often preferred to a gamma model because it is easier to interpret at the end.

GLM

Generalized Linear Models encompass a lot of models and can be used in a coherent manner, similar to linear regression, i for many types of outcomes:

- ▶ Numerical and real: linear regression (normal GLM)
- ▶ Counts: Poisson GLM
- ▶ Positive: log-linear regression or gamma GLM
- ▶ Binary: logistic regression (binomial GLM)
- ▶ Categorical: multinomial model
- ▶ $[0, 1]$ data (like %): beta regression.

and others.